

Research Question & Hypotheses (Revised): Trustworthy Measurement of Oral Soft Tissue in OCT

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In one sentence: we want a system that measures the thickness of the oral epithelium from OCT scans, says honestly when it *cannot* measure, puts a reliable uncertainty range on every number it gives, and learns this from very few hand-labelled scans — as the first building block of a future clinical decision support system (CDSS).

What changed from the first version. This version merges an external review of the plan. It removes “first-ever” claims that turned out to be untrue, makes the system’s ability to say “I can’t measure this” a central feature, uses more careful wording about attenuation and uncertainty, and keeps our physics-informed learning idea as the method that makes the system work with few labels.

1. The problem in plain words

What we want to measure. The lining of the mouth (oral mucosa) is built in layers. On top is the **epithelium**, a sheet of cells. Below it is the **lamina propria**, connective tissue full of collagen and blood vessels. The line where they meet is called the **epithelial-stromal boundary (ESB)**. Many conditions change these layers: the epithelium can thicken or thin, and the ESB can blur or disappear. Today clinicians judge this by eye and confirm it with a biopsy.

Why OCT. OCT takes cross-section pictures of the top 1–2 mm of tissue without cutting or radiation. In OCT images the epithelium looks darker and the lamina propria brighter, so the boundary between them is visible. Earlier studies used OCT to measure normal epithelial thickness at different sites in the mouth and found it varies a lot by site — about 100 μm on the floor of the mouth versus about 660 μm on the cheek ([Di Stasio et al., 2019](#)). So “thick” or “thin” only means something relative to the site.

Why this is hard.

- **Data is scarce and guarded.** We did not find an openly downloadable oral OCT dataset with clinical labels. The important datasets sit with individual research groups and are available only through collaboration. (This is a statement about access, not proof that no dataset exists anywhere.)

- **Labels are expensive.** Tracing boundaries by hand takes expert time.
- **Scans vary in quality.** Motion, saliva, poor focus, the probe pressing on soft tissue, or shadows can hide the boundary. A system that still outputs a number in those cases is dangerous.
- **A clinical tool must be honest about its limits.** It needs to say how sure it is, and when it cannot measure at all.

What we will build first. Not a diagnosis tool. A **measurement layer**: something that finds the tissue surface and the ESB, reports thickness in micrometres, flags when the ESB cannot be seen, and gives an uncertainty range for each measurement. A future CDSS would sit on top of this.

2. What others have done, and what is left for us

Existing work we build on

- **Automatic boundary finding already exists.** A group in Vancouver built a deep-learning pipeline that checks the field of view, detects artefacts, finds the tissue surface, and finds the ESB, producing maps of epithelial depth across healthy, precancerous and cancerous tissue ([Hill et al., 2024](#)).
- **Attenuation-based measurements already exist.** The same group then measured shape, ESB visibility, and how fast the signal fades with depth (attenuation) as markers of dysplasia and cancer ([Malone et al., 2024](#)).
- **Learning from few labels already exists — including skin-to-oral transfer.** Liao and colleagues used semi-supervised learning to segment and denoise healthy oral OCT, including labels transferred from skin OCT ([Liao et al., 2025](#)).
- **Normal thickness values by site** have been published ([Di Stasio et al., 2019](#)), and earlier work examined the epithelium–lamina propria boundary in oral precancer (Lee et al.).
- **Physics-aware deep learning exists in other OCT areas:** recovering attenuation in heart-artery OCT ([QOCT-Net, 2023](#)) and using simulated physics to help retinal OCT models ([PhysioSpeck-Net, 2026](#)).

Claims we must not make

- “First oral OCT segmentation method.”
- “First epithelial thickness measurement in oral OCT.”

- “First attenuation-based oral OCT analysis.”
- “First low-label oral OCT method” or “first skin-to-oral transfer.”
- “No oral OCT dataset exists.”

What is still open — our contribution

No study we found combines all of the following for oral soft tissue (this is based on our search, so it must be re-checked before submission):

1. An explicit **“cannot measure” output** when the boundary is not visible, instead of a forced guess.
2. A **quality check** that explains *why* a scan was rejected (motion, shadow, poor contact, etc.).
3. Thickness reported in **physical units** with the conversion stated.
4. A **physics-informed learning method** that lets the model learn from unlabelled scans, tested at small numbers of labelled *patients*.
5. **Calibrated uncertainty ranges** (conformal prediction) checked on patients the model never saw.
6. **Strict patient-level separation** of training, calibration, and test data, so results aren’t inflated.

The novelty is the **combination**, done rigorously — not any single piece.

3. The research question

Main question

With only a small, access-restricted set of in vivo oral OCT scans, can we build a system that measures epithelial thickness accurately, refuses to measure when the boundary can’t be seen, and gives an uncertainty range that is reliable — while needing far fewer hand-labelled patients than a standard model?

This is deliberately a question about **measurement**, not diagnosis. It can be answered with labelled OCT volumes and scan information alone, without needing a large set of biopsy-confirmed cancers.

Sub-questions

- **Q1 — Quality.** Does telling the model about scan conditions (depth, focus, signal strength, visible artefacts) and letting it say “cannot measure” reduce errors in difficult scans, without making it worse on normal ones?
- **Q2 — Few labels.** Does our physics-informed learning method keep accuracy high when only a few patients are labelled, compared with a standard model and with an existing semi-supervised approach?
- **Q3 — Trust.** Do the uncertainty ranges contain the true thickness as often as promised on unseen patients, and are they narrow enough to be useful?
- **Q4 — Transfer (optional).** If we are allowed to use OCT data from other tissues, does pretraining on it help, and does a physically similar tissue (oesophagus) help more than a large but different one (retina)?

4. Hypotheses

For each hypothesis: what we expect, how we test it, and what result would prove it wrong. Every comparison uses the same patients, the same network size, and the same training rules — only the thing being tested changes.

H1 — A quality-aware model makes fewer dangerous errors (main hypothesis)

- **What we expect.** A model that (a) receives information about each scan — depth of the boundary, distance from focus, local signal strength, and artefact flags — and (b) can output *measured*, *uncertain*, or *cannot measure* will place the ESB more accurately and produce fewer false measurements in difficult scans (deep, faint, shadowed, or blurred boundaries), compared with an identical model without these features. Its accuracy on ordinary scans should not get worse.
- **How we test it.** Decide in advance what counts as a “difficult” scan. Compare the two models on: ESB position error in micrometres, thickness error and bias, and how often each correctly says “cannot measure” when experts also could not see the boundary.
- **Proves it wrong if** the quality-aware model is not better in difficult scans, or is worse on ordinary scans.

H2 — Physics-informed learning needs fewer labelled patients

- **What we expect.** Our method teaches the model to *redraw* each scan from its own description of the tissue (layer positions and how fast the signal fades in each layer), using the known physics of OCT. Because this needs no labels, every unlabelled scan helps. We expect it to beat, at small label budgets, both (a) the same model trained only on labelled scans and (b) a strong semi-supervised method trained on the same data.
- **How we test it.** Label budgets are counted in **patients**, not images: for example 5, 10, 20 and all training patients. Each budget is repeated with different random patient choices, and results are reported with uncertainty.
- **Proves it wrong if** there is no clear improvement over both comparisons at the small budgets.

H3 — The uncertainty ranges are reliable and useful

- **What we expect.** Using **conformal prediction** (a statistical method that sizes uncertainty ranges from the model's errors on held-out cases), the 90% ranges will contain the experts' thickness in about 90% of new patients, and will be narrow enough to be clinically meaningful — a usefulness limit we set in advance.
- **Important wording.** This is an *average* guarantee over patients like those used for calibration. It does not promise every individual range is right, does not automatically hold for each mouth site separately, and does not carry over to a new device or scanning protocol without recalibrating ([Angelopoulos & Bates, 2021](#)).
- **How we test it.** Freeze the model first. Calibrate on one group of patients; check on another group never used before.
- **Proves it wrong if** coverage falls clearly below 90%, or the ranges are too wide to be useful.

H4 — Transfer from other tissues (optional, only if data access allows)

- **What we expect.** Starting from a model pretrained on OCT of another tissue will help compared with starting from scratch. We will compare oesophagus OCT (a similar lining tissue), skin OCT, retinal OCT of similar size (different tissue), ImageNet, and pretraining only on our own unlabelled oral scans.
- **Not assumed in advance.** If a much larger source wins, the conclusion is that size mattered — not that tissue similarity is irrelevant.
- **Proves it wrong if** no pretrained source beats training from scratch at small budgets.

5. Scope: what the first paper will and won't claim

The first paper will show	The first paper will not claim
How accurately the system finds the surface and ESB, and measures thickness, within our device and protocol	That it detects or rules out cancer or dysplasia
How reliably it says “cannot measure” and why	That it decides whether a biopsy is needed
Whether physics-informed learning reduces the labels needed	That it replaces histopathology
Whether its uncertainty ranges are reliable on unseen patients	That it works on any device (“device-independent”)
How repeatable measurements are between scans	That its confidence is “guaranteed” for each patient

A note on attenuation. How quickly the OCT signal fades with depth depends on the tissue, but also on the device, focus, and image processing. So we call it a **signal-decay measure specific to our system and protocol**, not a pure tissue property, and only compute it inside layers where the signal is good enough.

Where this leads. The measurement layer is the first step toward a CDSS. Later studies would link the measurements to clinical findings, test the tool with clinicians in “silent mode” (running alongside care without influencing decisions), and only then study its effect on care.

Suggested titles (avoiding words like diagnosis, cancer detection, or device-independent):

- *Quality-aware, physics-informed measurement of oral epithelial thickness from OCT with calibrated uncertainty*
- *Learning to measure oral mucosa in OCT from few labels, and knowing when not to*

6. Risks and what to confirm first

Risk	Why it matters	What we do
No access to a device or in vivo scans	Everything depends on it	Confirm with the project lead; healthy volunteers are the fastest start
Only processed (display) images available	Physics-based parts need raw signal	Ask partners for raw or minimally processed data; without it, keep physics claims modest
Too few patients	Results would not hold up	Start with a small pilot (planning estimate: about 10–15 patients) to check feasibility before the main study
Only one annotator	Can't tell how hard the task is for humans	Two independent experts on a shared subset
Probe pressing on soft tissue	Changes measured thickness	Record contact and pressure; standardise the scanning protocol
The simple physics model doesn't fit real tissue well	Weakens H2	Add a small learned correction; test the method on simulated scans first
Someone publishes a similar combination	Novelty	Re-check the literature just before submission

Timeline reality. A careful study (pilot, then a main study with separate training, calibration and test patients) takes months, not weeks. What can be ready this week is the direction, the hypotheses, and the plan.

7. Words to know

Term	Plain meaning
Epithelium	The top layer of cells lining the mouth
Lamina propria (stroma)	The connective tissue under the epithelium
ESB (epithelial–stromal boundary)	The line between the epithelium and the lamina propria
A-scan / B-scan / volume	One depth line / one cross-section image / a stack of cross-sections
Axial thickness	Thickness measured straight down along the scan direction
Refractive index	Number used to convert OCT's optical depth into real depth (about 1.38 for oral soft tissue)
Attenuation / signal decay	How fast the OCT signal fades with depth
Quality gate	A check that rejects unusable scans and says why
“Cannot measure” state	The model's honest output when the ESB isn't visible
Label budget	How many patients have hand-drawn boundaries used for training
Semi-supervised learning	Learning from a mix of labelled and unlabelled data
Physics-informed learning	Training that uses known physical laws (here, how OCT signals form) as a teacher
Pretraining / transfer	Starting from a model already trained on other data
Conformal prediction	A statistical way to size uncertainty ranges so they contain the true value a chosen share of the time
Coverage	The share of cases where the true value fell inside the range
Calibration set	Patients kept aside only to size the uncertainty ranges

Term	Plain meaning
Patient-level split	Keeping all scans from one person in the same group (training, calibration or test)
CDSS	Clinical decision support system: software that helps clinicians decide, without deciding for them

8. References

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References without links were cited in the external review and should be checked before use.